

Adam Al Jaaouni

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EDUCATION

University of Waterloo

Bachelor of Computer Science (Co-op)

- Specialization in AI & SWE

Waterloo, ON

Sept. 2025 – Aug, 2029

TECHNICAL SKILLS

ML/CV: YOLOv8/v11, Core ML, OpenCV, Vision, Ultralytics

Languages: Java, Python, C, C++, Swift, TypeScript, SQL, Racket

Frameworks: PyTorch, React, React Native, Node.js/Express, SwiftUI/UIKit

Tools: Git/GitHub, Docker, PostgreSQL, Claude Code

EXPERIENCE

Computer Vision Engineer Intern

Gametime Technologies

Waterloo, ON

May 2026 - Present

- Built a **React Native + native Swift** iOS app for live sports broadcasting across **17 sports**, with real-time overlays, multi-destination streaming, and **on-device auto-scoring**.
- Built an end-to-end basketball computer vision pipeline, taking a fine-tuned **YOLOv8** detector from **PyTorch training** through **Core ML/Vision deployment** for live on-device inference at **~12 FPS**.
- Trained a **5-class YOLOv8** detector via **transfer learning**, achieving **93% mAP50** (0.86 precision / 0.90 recall), then optimized inference with a cropped second pass to recover distant ball detections.
- Designed a **multi-frame state machine** to distinguish rim bounce-outs from net-occluded makes, replacing a brittle single-frame make-detection heuristic.
- Improved **1v1 player separability** from **0.41 to 0.85** using a lightweight **YCbCr shirt-color descriptor**, avoiding the need for a deep re-identification model.
- Maintained **feature parity** across **Swift, TypeScript, and Python** implementations, validated with a Python/OpenCV integration harness and **250+ Jest unit tests**.

Team and Software Captain, Robot Driver/Coach

FRC Robotics Team 3739

London, ON

Sept. 2022 – Apr. 2025

- Engineered multi-DOF robotic arms, elevators, and flywheel shooters via **PID motor, pneumatic, and PWM/servo control**.
- Built **AprilTag CV** and **neural network target detection** for autonomous navigation, achieving **±2cm** accuracy via **PID control** fused with vision/encoder odometry.
- Led team to **provincial division finalist (3x)** and **2025 World Championships**; grew Programming Team from 2 to 12 members.
- **Dean's List Semi-Finalist**; taught **Java, OOP, and Git** fundamentals.

PROJECTS

Ping Pong Ball Object Detection using YOLOv11s

Jan. 2026

- Trained a custom **YOLOv11s** object detection model GPU accelerated training with **Google Colab**.
- Curated/labeled YOLO-formatted datasets with **training and validation** splits for robust generalization.
- Evaluated model using mAP@0.5, mAP@0.5:0.95, precision, and recall to assess generalization.
- Implemented Python-based inference pipelines using said YOLO model for real-time object detection.

Full Stack News Aggregator Web App

Jan. 2026

- Built a full-stack application using **React and Node.js/Express** with real-time **NewsAPI** integration.
- Implemented **category and keyword search**, secure environment variables, and optimized API handling.
- Used Docker to containerize and then AWS ECS to manage and deploy the containerized web app.